Sequence Message	Description
Drive fault during speed 1 tuning	Indicated if there is a fault during the auto-loss sequence. Refer to the DRIVE diagnostic screen
Auto-loss tuning complete	Indicates that the sequence has been completed
Drive not running	Indicates that the drive is not running
Drive failed to reach zero speed	Indicates an issue when the set point has been set to zero at any stage in the sequence, but the motor has not reached it within the specified time. Contact BML for further advice
Auto-loss compensation not enabled	Indicates that there is an machine interlock that is stopping the auto-loss sequence being started
Vessel not closed – unable to start	Indicates that the sequence cannot be started, because the chamber is not closed
Internal/external E-stop active	Indicates that the sequence cannot be started, because of an external e-stop or G-stop condition.
Unwind/rewind diameter exceeds safe limit for auto-losses	Indicates that the sequence cannot be started, because either the unwind or rewind have a large diameter roll on them which will not allow the sequence to start

Table 11.4-2 Sequence Message Descriptions

After completion of the first set of auto loss procedures, it is suggested that the friction curves are saved as reference curves for further reference.

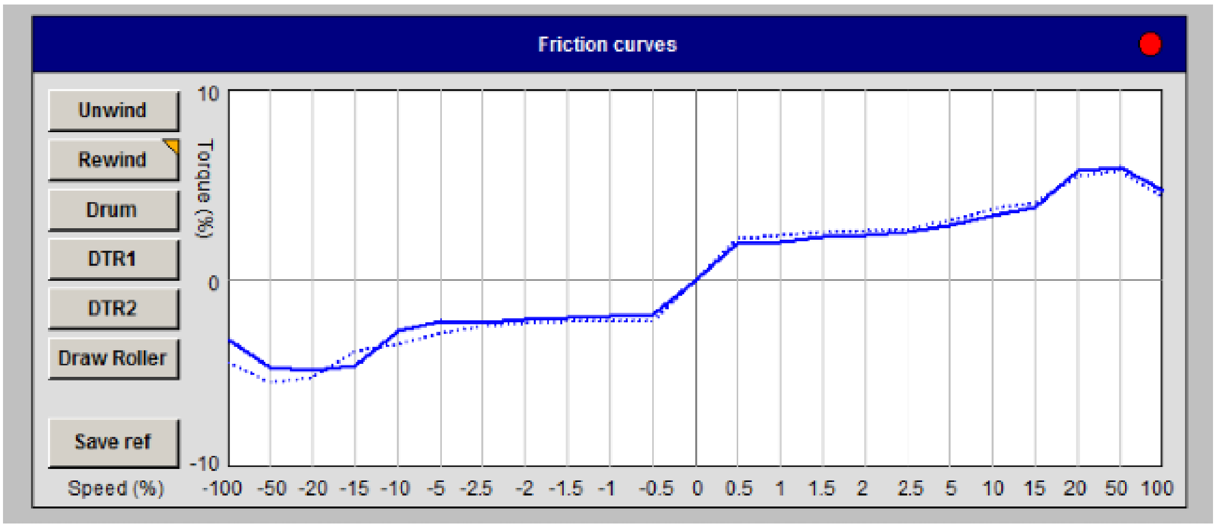


Figure 11.4-2 Auto-loss HMI “Friction Curve” Section

11.5 Rewind Layarm Alignment

This chapter describes the alignment procedure for the rewind layarm.

Rewind Layarm

The purpose of the rewind layarm is to guide and support the film to the rewind roll.

If the final roller on the layarm mechanism is not parallel to the rewind roll, winding defects can be created.

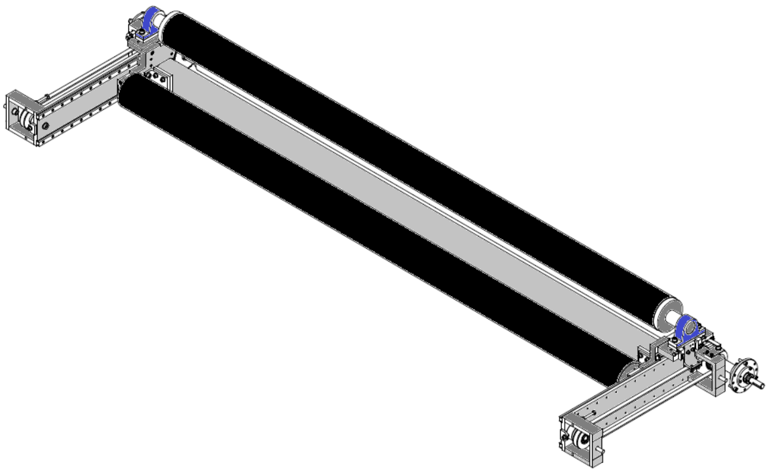


Figure 11.5-1 Overview drawing of rewind layarm assembly.

To check mechanical alignment between the last roller in the layarm assembly to the rewind roll, the following procedure is suggested:-

1. Place a reel bar in the rewind chuck, without any core on it allowing the bare metal surface to be used.
2. Use the rewind “IN” push-button to drive the layarm to a fixed distance from the reel bar.
3. Disable the drive system by switching the G-stop isolator off and “lock” it out
4. Use mechanical "stick micrometre" to measure the gap at both ends of the gap
   An alternative method is to use a “pi-tape”

   Figure 11.5-2 Photograph of “stick micrometres”
5. Refer the mechanical assembly drawing.
6. Modify the distance, if required, by adjusting the belt tensioners on either side. This is achieved by loosening off the nuts (S20) to move the “threaded rod” (S014) by tightening one of the nuts on the “threaded rod”, then tightening the other nut.

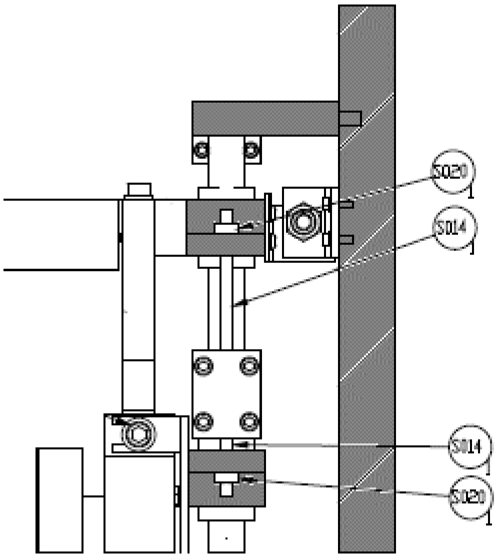


Figure 11.5-3 Drawing of adjustment components

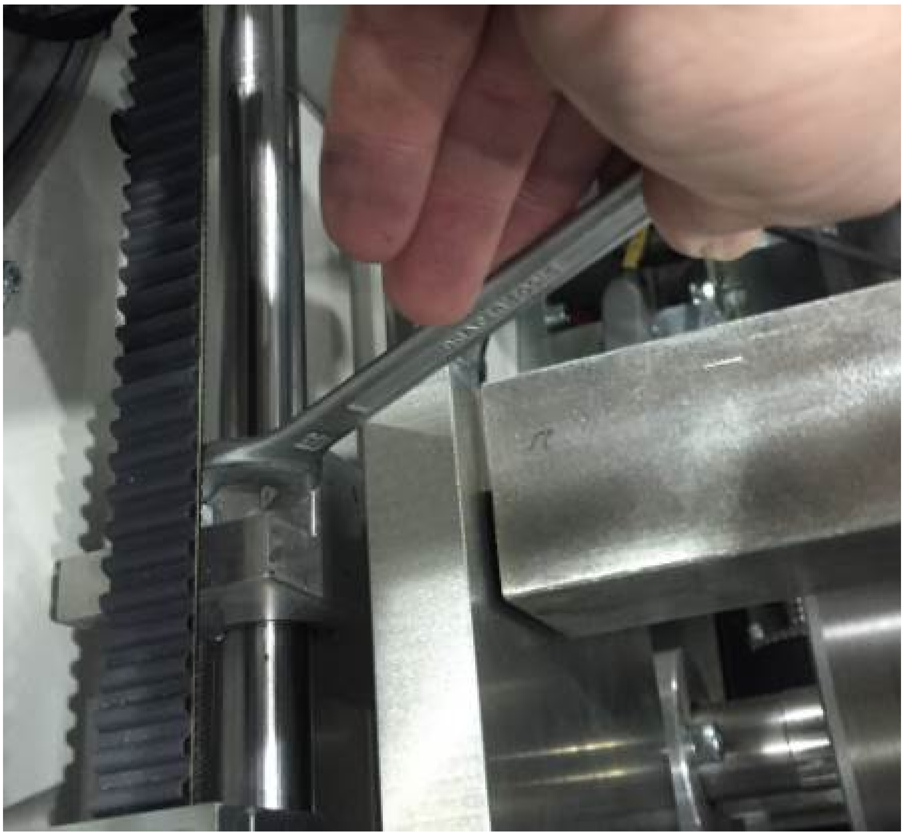


Figure 11.5-4 Example of a technician adjusting the position

If the adjustment cannot be made using the available adjustment through this method, the clamp may need to be readjusted on the drive belt.

The clamp is released by loosening the four bolts. This will allow the side of the assembly to be moved on the slide rails, before re-clamping.

Figure 11.5-5 Example of a technician releasing the clamps on the belt

7. Adjust bolts (S16) to move the position of the bracket, checking the distance between the roller and the reel shaft using a “stick micrometer”, by the use of the jacking screw.

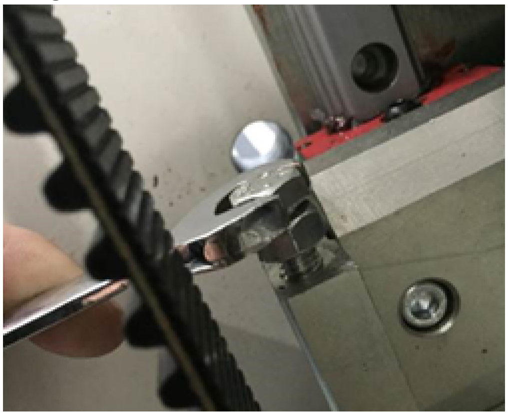


Figure 11.5-6 Example of a technician adjusting the jacking screw.

The distance should not normally change under normal operation; however, it has been found that incorrect loading/unloading procedures can cause the belt to jump the sprockets.

Belt Tensioning

Ensure that the positioning belts have even tension on both sides of the positioning bracket.

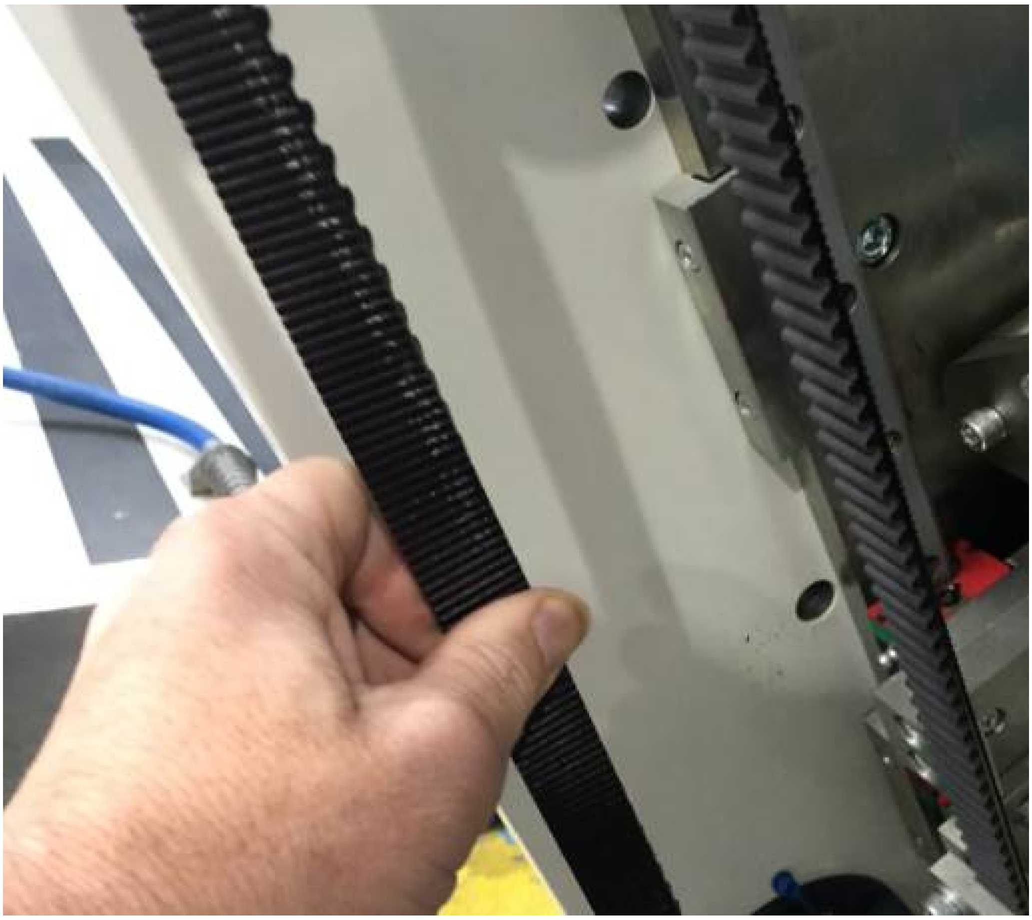


Figure 11.5-7 Example of a technician checking the belt tension.

Replacement of Timing Belts

To replace the belts, the drive shaft has to be supported while the outboard plate support bearing is removed to allow a belt to be passed over it.

Assuming that belt has broken:-

1. Move layarm to safe position or support it
2. Loosen Belt Clamp and remove
3. Loosen belt tensioner assembly.
4. Support Drive shaft
5. Remove Cartridge bearing
6. Pass new belt through hole in side plate and over drive shaft
7. Replace cartridge bearing
8. Install belt tensioner and tension up belt
9. Realign the layarm assembly

	Warning !
	Lockout the movement isolator and apply lock before attempting to work in this area. **Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.**

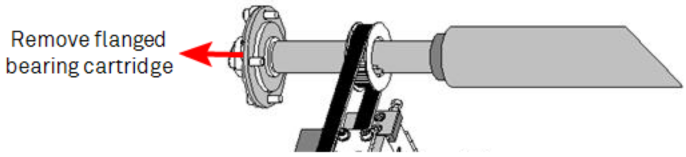


Figure 11.5-8 Layarm belt drive shaft

Assuming that Belt is damaged:-

10. Move layarm to safe position or support it
11. Loosen Belt Clamp and remove
12. Loosen belt tensioner assembly.
13. Support Drive shaft
14. Remove Cartridge bearing
15. Cut existing belt and remove
16. Pass new belt through hole in side plate and over drive shaft
17. Replace cartridge bearing
18. Install belt tensioner and tension up belt
19. Realign the layarm assembly

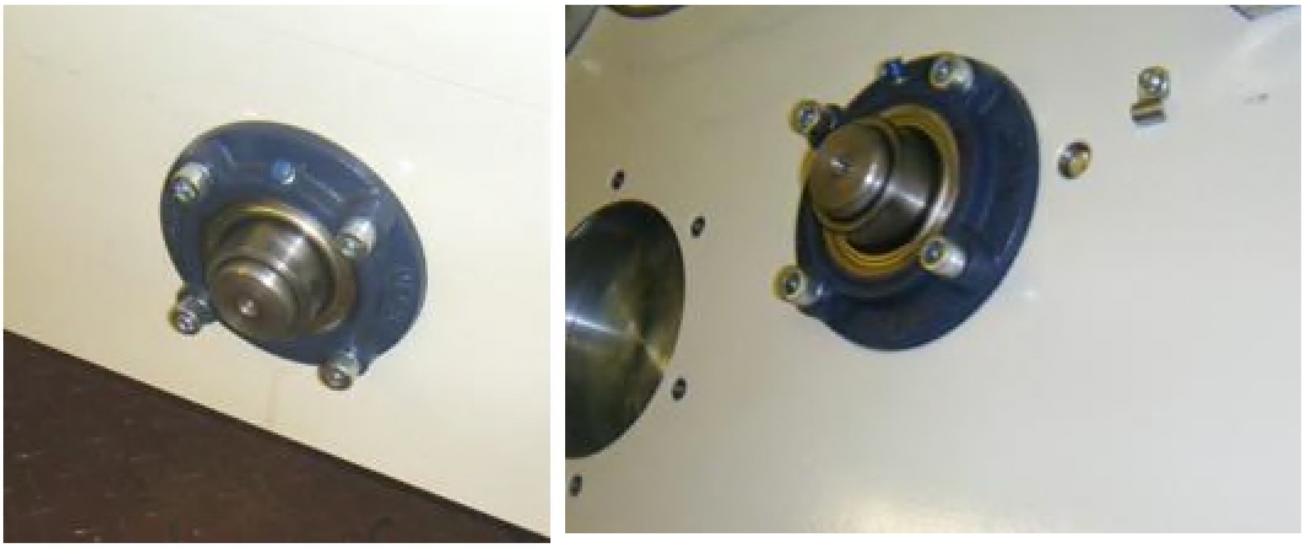


Figure 11.5-9 Layarm belt drive flanged bearing cartridges

Layarm Drive Set Up and Calibration Procedures

The layarm position calibration is set at the Heywood facility and generally in the field will not require modifying under normal operational conditions.

However, under the following circumstances, the position calibration can become non- accurate.

- Repositioning of “home” limit Switch
- Jumping of external or internal drive Belts
- Repositioning of layarm “Home” switch striker plate

This will be indicated by incorrect diameter sensing at the start of the roll, with tensions levels too high or low, leading to web breaks, or layarms too close to or far away from the rolls.

	Note !
	Whenever carrying out maintenance procedures on the machine that relates to the layarm limit switches, ensure that there position is identified to allow accurate repositioning.

Calibration

All other calibrations are carried out in the "Maintain machine configuration" screen.

This can be accessed at the customer maintenance engineer level.

Figure 11.5-10 Maintain Machine HMI screen

Parameter	Explanation
Layarm Inch Torque (Bits)	The amount of torque normally used to raise of lower the layarm using the local "In and Out" buttons
Layon Null (Bits)	This is the balance torque value that the layarm requires, before any movement takes place in the layon mode.
Span (mmx13)	The distance between the shafted core position and home position in mm x13

Table 11.5-1 Layarm calibration parameters

Span Calculation:

To calculate the 'Span' for the full movement, consideration must be taken of the minimum core diameter, which has been previously entered in machine software configuration. On standard shafted machines, this is 165mm

Span = Distance between the shafted core position and home position in mm.

This distance must be physically measured as accurately as possible in millimetres

This millimetre value is multiplied by a value of 13 and is then entered into the 'Span' into the box on the screen.

Layon Null

To find the 'Layon Null' value, the following procedure should be used

Using the raise and lower rear pushbuttons, move the layarm assembly to a mid-movement position on the slide.

Reduce the 'Layarm inch torque setting' parameter value, on the on the HMI screen.

Then use the raise button to energise the motor. By adjusting to different values, it will be found that the layarm will at a value not move either up or down, but with a gentle push by a finger will move it.

This is the value that should be entered into the 'Layon Null' value parameter on the on the HMI screen.

Ensure that the ‘Layarm inch torque setting’ parameter value is reset to its previous value.

Use the Save button to load the values to the PLC. Failure to do this will mean that when power to the machine is removed, the values will be lost.

11.6 Motorised Spreader Roller Replacement & Position Feedback

These instructions should be used, if “Kickert” manufactured bow spreader roller has been replaced. It will have an orange cylindrical body, containing a feedback potentiometer. There are other Kickert spreader rollers with an external remote potentiometer. These are covered by a separate write up.

Figure 11.6-1 Kickert spreader roller

1. Remove the power source (circuit breaker) feeding the circuit/motor.
2. Lock out the traverse movement isolator or the main incoming supply isolator to make the machine safe.
3. Disconnect the wiring from the motor, noting where individual cores are connected to.
4. Mechanically replace the roller.

Caution !

Ensure that the roller is lifted as per the guidelines in the manual supplied with the replacement roller

5. Replace the wiring onto the motor as previously disconnected.
6. Apply the main power back to the machine, but not the supply to the motor at this time.
7. At the maintenance level of access, go to the spreader rolls section of the maintenance HMI screen.

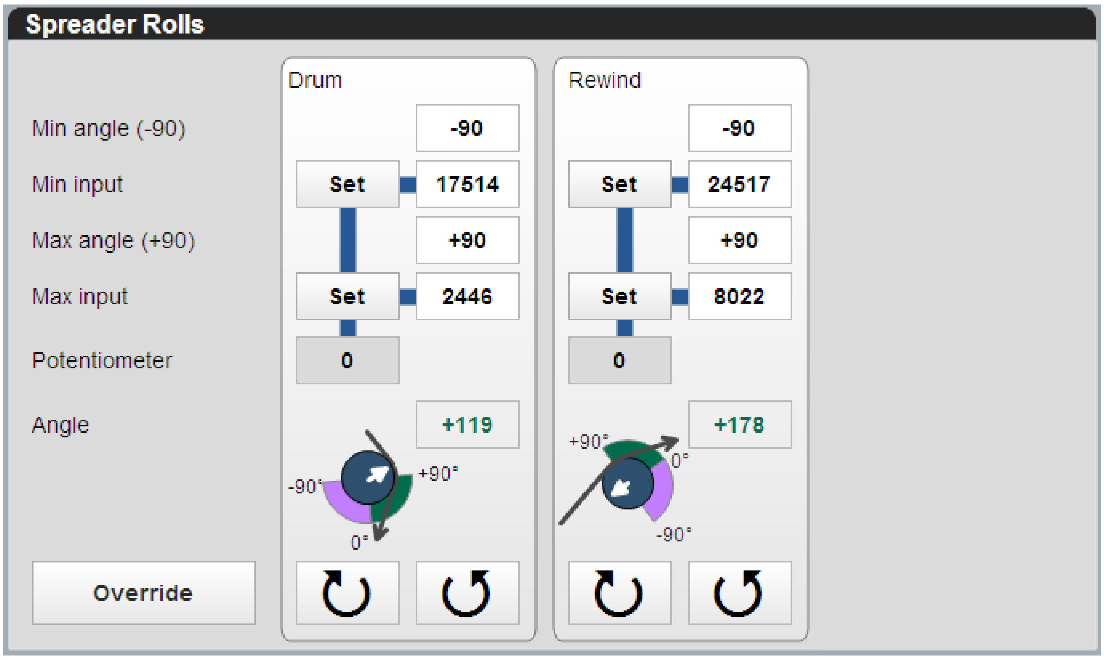


Figure 11.6-2 Spreader rolls section

8. Locate for the apex indicator on the end of the spreader roller.

Note !

Note Drum refers to the pre-drum spreader while the rewind spreader refers to the post DTR2 roller, not the one on the rewind layarm.

Figure 11.6-3 Kickert spreader roller apex indicator

9. Select the override button on the set-up screen to allow control from this location.
10. Using the “+” or “-“ button, move the roller apex on the roller to the zero degree position.
11. When at position, select the “SET” button for the “min input”
12. Using the “+” or “-“ button, move the roller apex on the roller to the 180 degrees position.
13. When at position, select the “SET” button for the “max input”
14. Remove the override button on the set-up screen to allow control from the drives - spreader interface screen.
15. Use the angle control on the rewind spreader angle section of the drives - spreader screen to change the set angle and monitor for the response for correct direction and feedback.

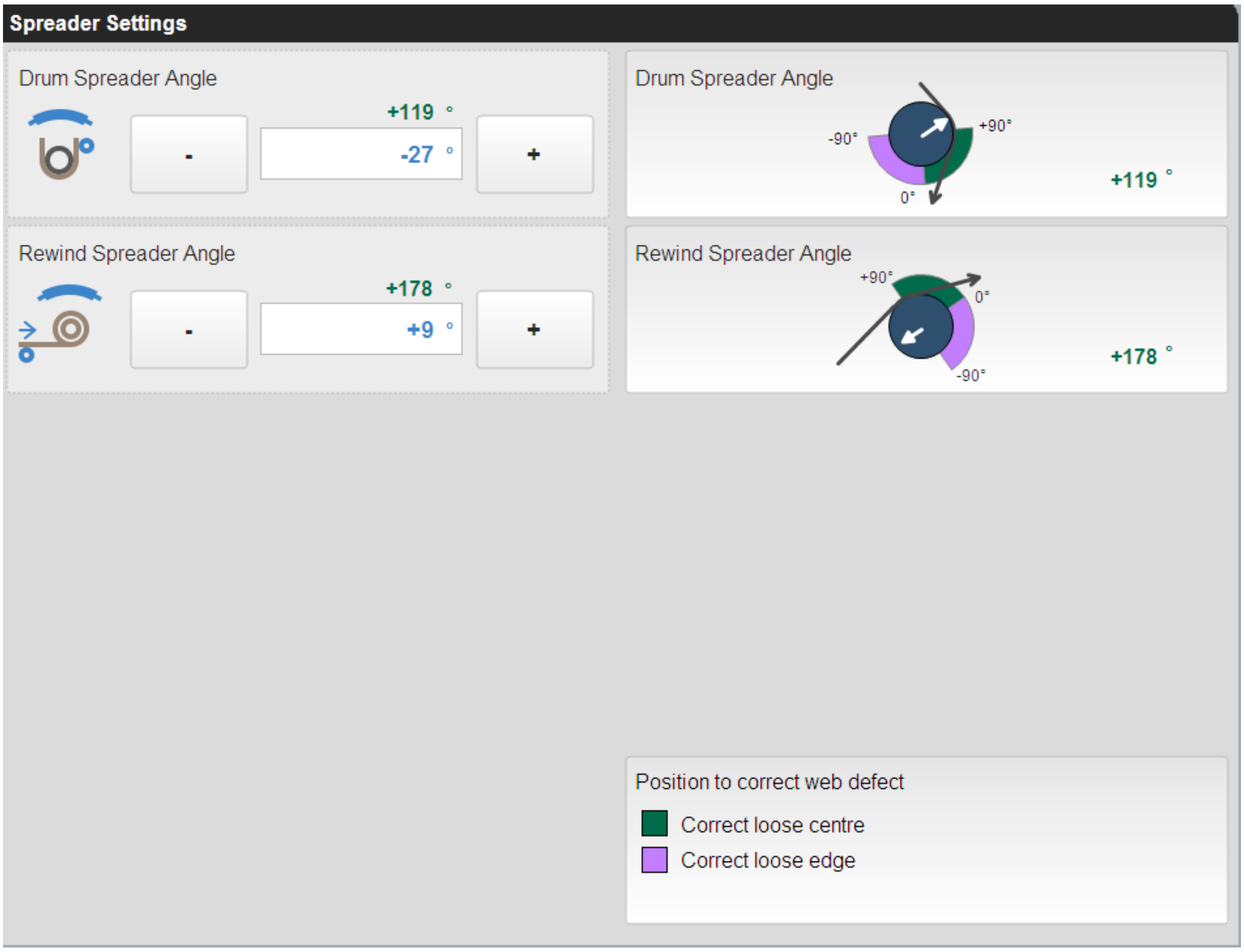


Figure 11.6-4 Rewind spreader section of drives HMI screen

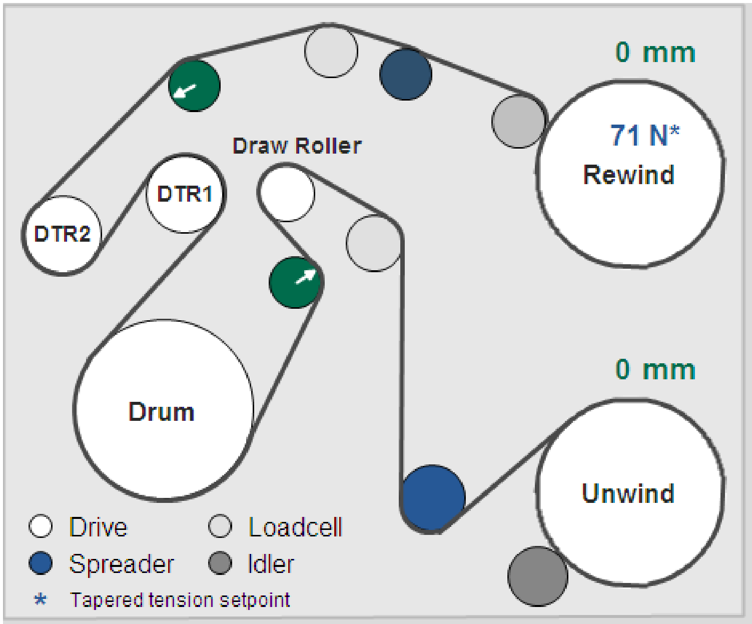


Figure 11.6-5 Drive screen indicating spreader roller feedback.

11.7 Unwind Layarm Roller Mechanical Alignment

This chapter describes the mechanical alignment procedure for the unwind layarm roller.

Unwind Layarm

The purpose of the unwind layarm roller, is as the following descriptions

- When in the contact mode, it will stop the upper layers of the substrate from telescoping (sideways movement)
- In the case of a web break situation, it will move up to the roll to stop it unwinding.
- Is used to make contact and give feedback on the roll diameter.

If the roller is not aligned parallel to the roll, winding defects can be created.

The assembly is positioned upon a linear slide arrangement, which travels towards and away from the unwind roll.

The assembly is moved by a timing belt/sprocket/shaft mechanism.

The shaft that moves the assembly, passes through the vacuum faceplate wall, through a rotary vacuum leadthrough, and is then attached to a motor.

The motor moves at a very accurate speed in either direction and is controlled from a drive interface unit.

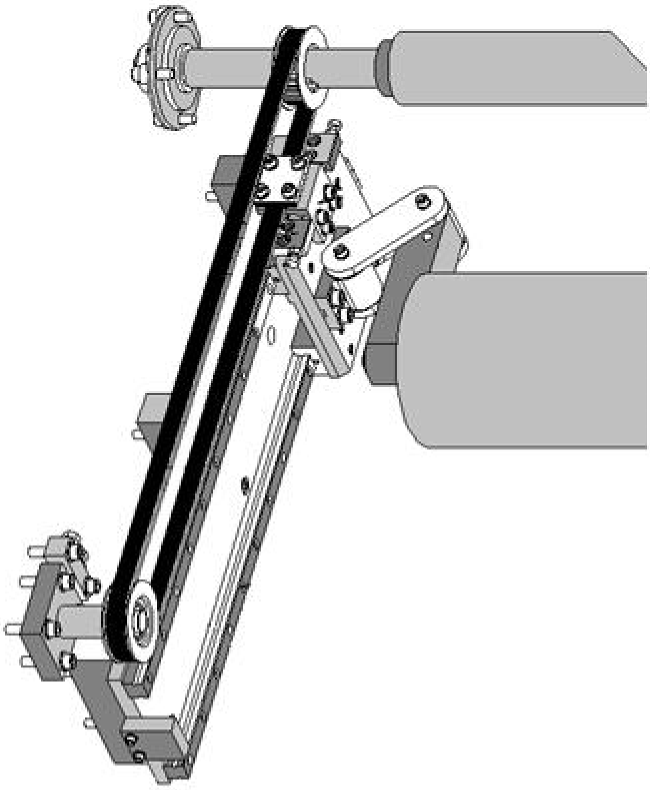


Figure 11.7-1 Unwind assembly

As the roll diameter increases, the layarm assembly will move away from it keeping the same gap.

Lay On (Contact) Mode

In this mode of operation, the arm is pressed against the roll.

The amount of force being applied is achieved by modifying amount of torque in the layarm motor.

Increasing the amount of torque in the layarm motor will increase the pressure.

This is normally set to a maximum value of 400 newtons. This is dependent upon the weight of the roller assembly for the rewind and the amount of torque available to the unwind

Note: 1KG = 13 newtons = 2.2 lbs

Movement Limit Switches/sensors

There are limit switches located at the fully retracted position (home), of the slide rails to de-energise the motor off when it touches the switch.

Note !

Incorrect setup or non-operation of the limit switches will corrupt the diameter set up function.

This may also lead to gaps not being correct when winding materials.

This switch is also used by the drive system to indicate that the layarm is in its “home” position for use, when calculating the location of the assembly, by use of the encoder.

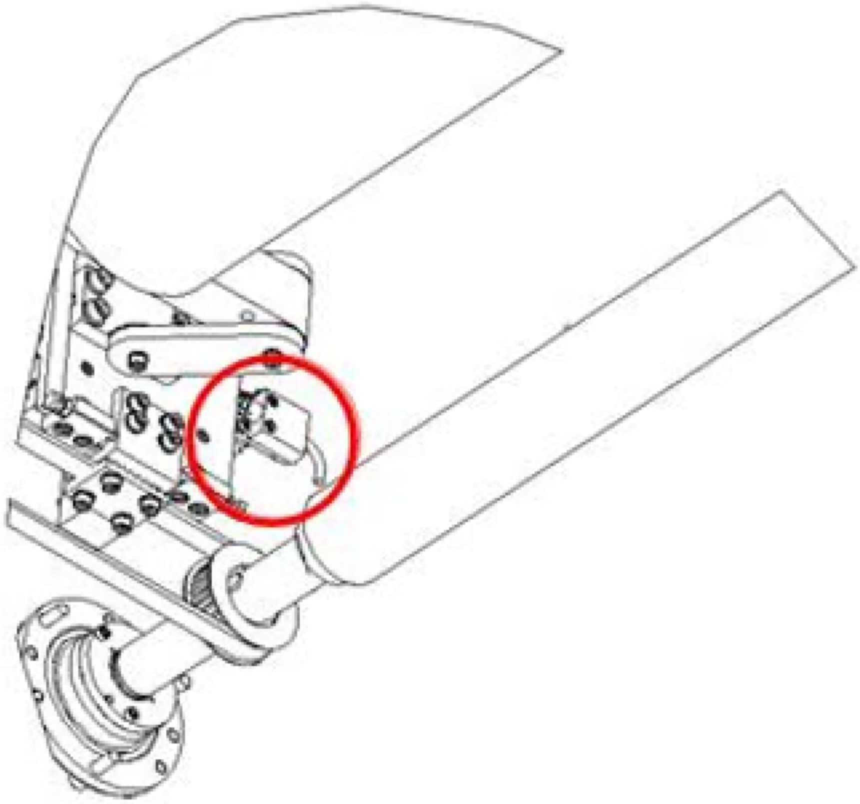


Figure 11.7-2 Unwind assembly showing limit switch position

Unwind and Rewind Layarm Drive Belt Tensioning/ Replacement

The rewind and unwind layarm assemblies are positioned from an external drive via an internal belt driven mechanism.

If these internal belts become loose, the position and alignment of the rollers will become affected.

Warning !

Lockout the movement isolator and apply lock before attempting to work in this area.

Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.

Unwind

The unwind belt tension can be adjusted by:

1. Loosen the shoulder screw, this will allow sliding movement of the adjustable sprocket.
2. Adjusting the tensioning bolt, this will tension or release the belt.
3. Tension up the belt as recommended in the relevant section of the vendor manual
4. Tighten the shoulder screw when the tension has been applied.

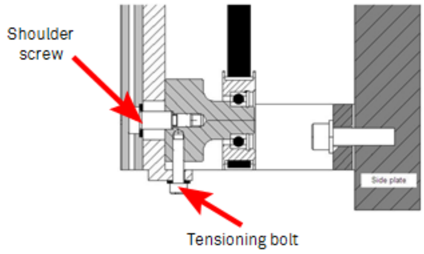


Figure 11.7-3 Unwind layarm drive belt tension adjuster.

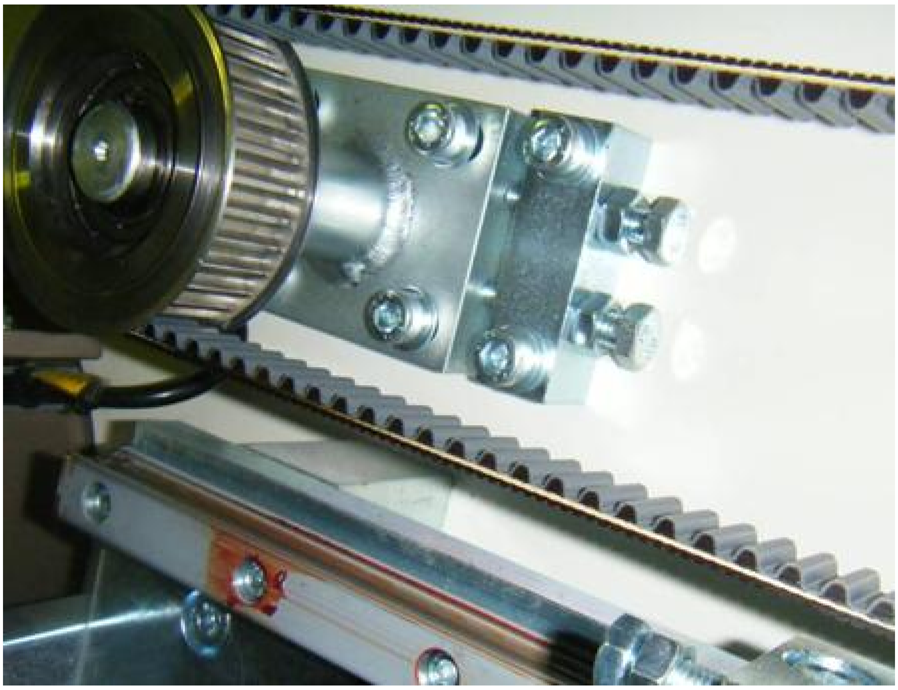


Figure 11.7-4 Unwind layarm belt tensioner

Replacement of Timing Belts

To replace the belts, the drive shaft has to be supported while the outboard plate support bearing is removed to allow a belt to be passed over it.

Assuming that belt has broken:-

1. Move layarm to safe position or support it
2. Loosen Belt Clamp and remove
3. Loosen belt tensioner assembly.
4. Support Drive shaft
5. Remove Cartridge bearing
6. Pass new belt through hole in side plate and over drive shaft
7. Replace cartridge bearing
8. Install belt tensioner and tension up belt
9. Realign the layarm assembly

Warning !

Indicates loss of life, severe personal injury, or substantial machinery damage will result if proper precautions are not taken.

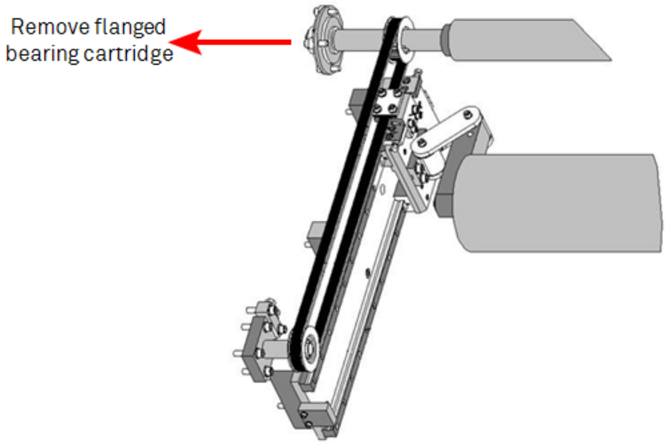


Figure 11.7-5 Layarm belt drive shaft

Assuming that Belt is damaged:-

10. Move layarm to safe position or support it
11. Loosen Belt Clamp and remove
12. Loosen belt tensioner assembly.
13. Support Drive shaft
14. Remove Cartridge bearing
15. Cut existing belt and remove
16. Pass new belt through hole in side plate and over drive shaft
17. Replace cartridge bearing
18. Install belt tensioner and tension up belt
19. Realign the layarm assembly

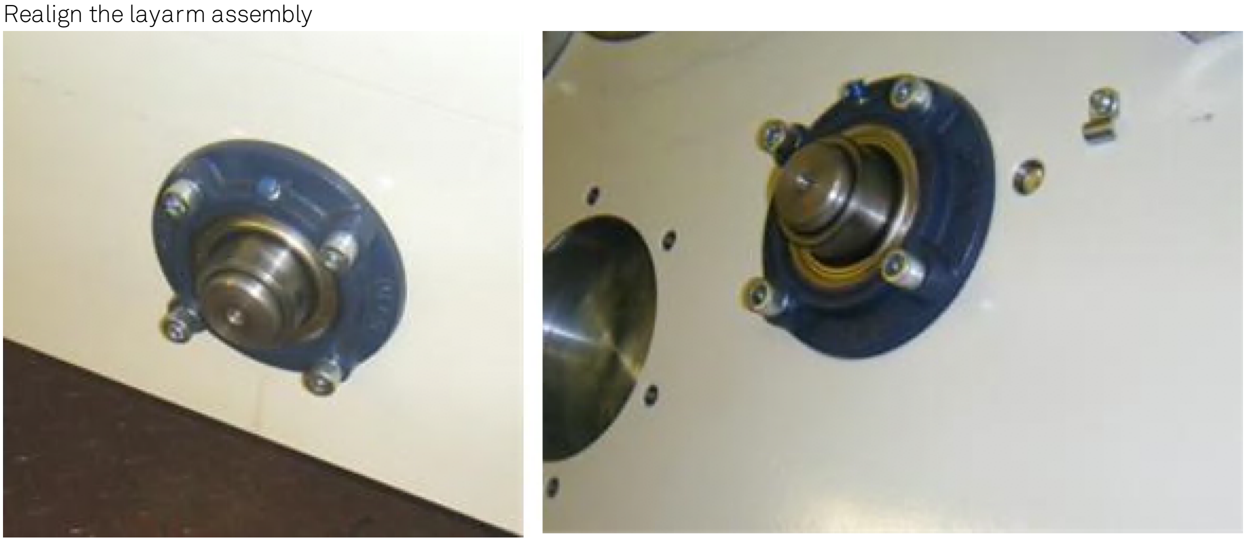


Figure 11.7-6 Flanged bearing cartridges for the layarm belt drive

Layarm Drive Set Up and Calibration Procedures

The layarm position calibration is set at the Heywood facility and generally in the field will not require modifying under normal operational conditions.

However, under the following circumstances, the position calibration can become non- accurate.

- Repositioning of “home” limit Switch
- Jumping of external or internal drive Belts
- Repositioning of layarm “Home” switch striker plate

This will be indicated by incorrect diameter sensing at the start of the roll, with tensions levels too high or low, leading to web breaks, or layarms too close to or far away from the rolls.

	Note !
	Whenever carrying out maintenance procedures on the machine that relates to the layarm limit switches, ensure that there position is identified to allow accurate repositioning.

Calibration

All other calibrations are carried out in the Maintain machine configuration screen on the HMI.All other calibrations are carried out in the Configure tab of the Maintenance page on the HMI.

This can be accessed at the customer maintenance engineer level.

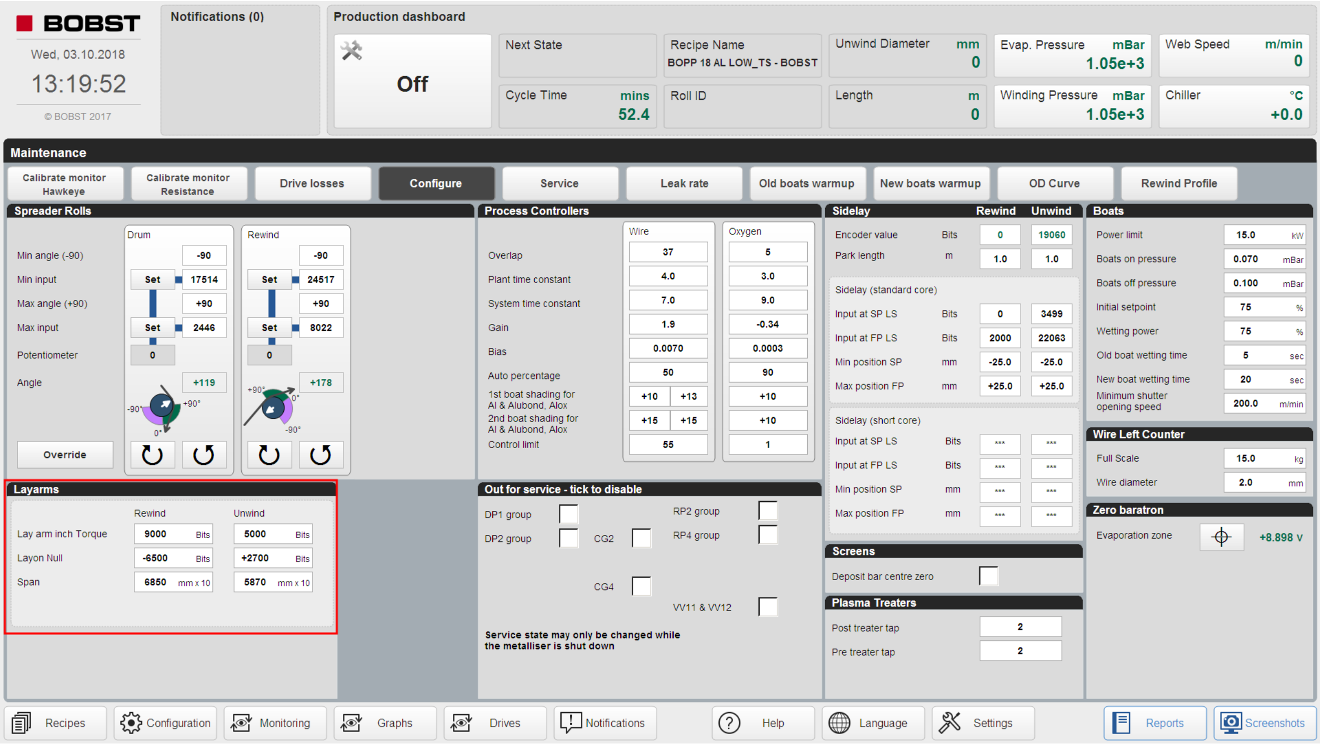


Figure 11.7-7 Layarms section of the configure tab in the maintenance HMI page

Parameter	Explanation
Layarm Inch Torque (Bits)	The amount of torque normally used to raise of lower the layarm using the local “In and Out” buttons
Layon Null (Bits)	This is the balance torque value that the layarm requires, before any movement takes place in the layon mode.
Span (mmx13)	The distance between the shafted core position and home position in mm x13

Table 11.7-1 Layarm Calibration Parameters

Span Calculation

To calculate the ‘Span’ for the full movement, consideration must be taken of the minimum core diameter, which has been previously entered in machine software configuration. On standard shafted machines, this is 165mm

Span = Distance between the shafted core position and home position in mm.

This distance must be physically measured as accurately as possible in millimetres

This millimetre value is multiplied by a value of 13 and is then entered into the ‘Span’ into the box on the screen.

Layon Null

To find the ‘Layon Null’ value, the following procedure should be used

Using the raise and lower rear pushbuttons, move the layarm assembly to a mid-movement position on the slide.

Reduce the ‘Layarm inch torque setting’ parameter value, on the on the HMI screen.

Then use the raise button to energise the motor. By adjusting to different values, it will be found that the layarm will at a value not move either up or down, but with a gentle push by a finger will move it.

This is the value that should be entered into the ‘Layon Null’ value parameter on the on the HMI screen.

Ensure that the ‘Layarm inch torque setting’ parameter value is reset to its previous value.

Use the Save button to load the values to the PLC. Failure to do this will mean that when power to the machine is removed, the values will be lost.

SECTION 12.0 PREVENTATIVE MAINTENANCE

This section provides information on preventative maintenance tasks in the following chapters:

To cover most aspects of preventative maintenance of the third party equipment supplied with this machine, manuals are provided, as supplied by the manufacturer. These should be referred to when preventative maintenance is undertaken on the machine.

These manuals are located on the documentation CD/DVD.

Warning !

Before any maintenance is carried out on any part of the rotating machinery, the machine must be brought to a stationary condition and be locked out.

Under no circumstances must a user attempt to clean the machine rollers when the mechanism is rotating or with tension is applied to the substrate.

Overhaul of any of the machines components must ONLY be carried out by qualified personnel who are fully conversant with all mechanical and electrical SAFETY INTERLOCK devices.

Maintenance personnel should read the suppliers manuals covering the components fitted to the machine before commencing any maintenance procedures on them.

Definition of Maintenance Personnel

Maintenance or service personnel will generally have a limited knowledge of the machine operation.

Warning !

Reduced level of safety.

Improper operation can cause serious injuries of the personnel and considerable equipment damage.

The maintenance and service personnel may operate the machine in normal operation, regularly perform necessary maintenance work and additionally perform all further service work which is necessary for proper operation. For this purpose, the machine may be operated in certain manual modes.

Maintenance of the machine comprises the following activities:

- Switching the system on and off.
- All work mentioned in the preventative maintenance schedule

Service of the machine comprises the following activities:

- Shutting down and dismantling the machine for access to the system components.
- Repair work on mechanical and electrical components of the machine.

Preventative maintenance and service of the machine may only be carried out by trained competent employees of the system owner.

For work on electrical components, training as an electrical technician or comparable technical training is recommended.

12.1 Preventative Maintenance Schedule

In an attempt to reduce downtime and consequent lost production through breakdowns, it is recommended that the checks (actions) detailed in the following schedule be fully complied with. Refer to the third part vendor manuals supplied by the original equipment manufacturers, for further information.

	Note !
	Failure to carry out scheduled maintenance of the equipment as described in the following pages will invalidate any warranty agreements.

Records of preventative maintenance procedures shall be kept and be available to the supplier of this equipment, Bobst Manchester Ltd. and their vendors, as part of the machine warranty.

This machine will report to the user when tasks are required to be carried out through the notification system.

To check on the talks to be done, the user can refer to the HMI MAINTENANCE – SERVICE screen. A explanation can be found in section 14.4 of this manual.

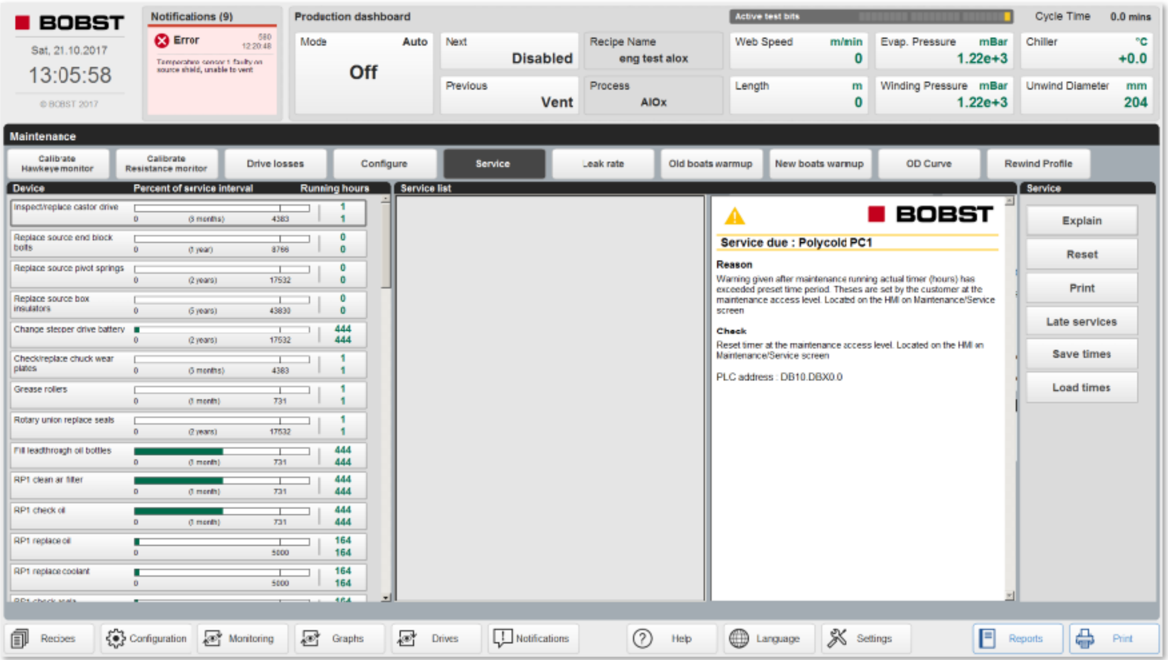


Figure 12.1-1 MAINTENANCE – SERVICES HMI screen

If there is a part of the machine that requires maintenance that is not referenced in this manual, contact Bobst Manchester Ltd. for advice.

This schedule is based upon a normal operation of 24 hours/day, 5 days/week. The frequency of preventive maintenance should be increased as the operation time increases. It is strongly suggested, that the equipment be maintained to the following schedule, even if the equipment is operated for fewer hours.

	Caution !
	Failure to carry out preventative maintenance on the machine will result in shortened life cycles of the parts supplied.

12.2 Preventive Maintenance Tasks

The following identified tasks should be carried out as indicated in the chart:

Warning!

Make sure safe lifting practices are used when any components are removed from the machine during maintenance.

Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.

After the diffusion pumps shut down allow enough time for any exhaust gasses to diffuse before any procedure is carried out.

Caution !

Failure to carry out maintenance on the machine will result in shortened life cycles of the parts supplied.

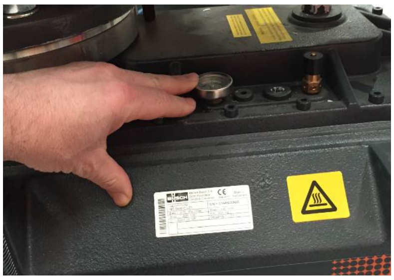
Not all the equipment listed in this section of the manual is applicable to this machine. Sections will only be included in the manual that are relevant to the machine supplied.

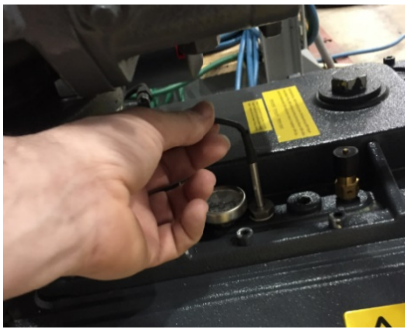
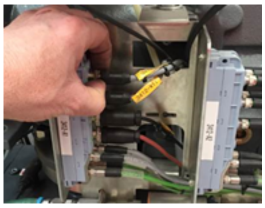
Section	Equipment
12.2.1 on page 563	Bush Screw Pump
12.2.2 on page 568	Aerzen Booster Pumps – Canned Motors
12.2.3 on page 571	Varian HS 35 Diffusion Pumps
12.2.4 on page 574	Vacuum Pipe work Isolation Valves
12.2.5 on page 576	High Vacuum Valve (VV7)
12.2.6 on page 582	Pneumatic (Air) Systems
12.2.7 on page 583	Vacuum Level Measurement Gauges
12.2.8 on page 585	Chamber Vacuum-Vent Valves
12.2.9 on page 587	Vacuum Systems O-rings
12.2.10 on page 589	Vacuum Pipeline Rubber Isolation Bellows
12.2.11 on page 590	Cryo-Generator – Brookes Automation
12.2.12 on page 593	Vacuum System Performance
12.2.13 on page 594	Vacuum Chamber
12.2.14 on page 595	GRE Chiller Unit (48-96KW Model)
12.2.15 on page 601	Source Viewing Flap & rotary viewing window assemblies
12.2.16 on page 604	MKS Mass Flow Controllers
12.2.17 on page 606	MKS Baratron Vacuum Gauge - Plasma Treater

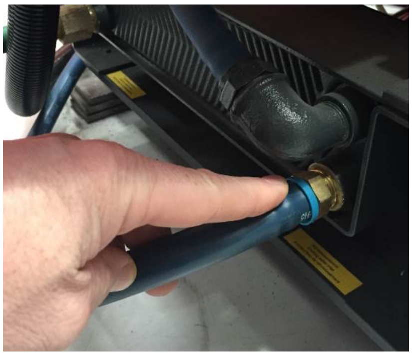
Section	Equipment
12.2.18 on page 607	HMI Operator Interface Screen
12.2.19 on page 610	Coating Shield and Shutter
12.2.20 on page 615	Aluminium Wire Feeder Units
12.2.21 on page 623	Inline Optical Monitor System
12.2.22 on page 625	Evaporation Source Area
12.2.23 on page 634	Winding Mechanism
12.2.24 on page 645	Chucking System & Reel Shafts – 152mm Standard
12.2.25 on page 654	Winding Cart Drive System Belt drive
12.2.26 on page 655	Winding Cart Cable Coiler
12.2.27 on page 657	Winding Cart Traverse Drive Motor/Gearbox Unit
12.2.28 on page 659	Drum Coating Shield/Zone Sealing
12.2.29 on page 662	External Cooled Rollers Rotary Unions – Liquid - Deublin
12.2.30 on page 664	Evaporation Source Electrical Busbar Systems
12.2.31 on page 668	Electrical panels & Air Conditioners - Rittal
12.2.32 on page 671	Pre Plasma Treater – Planar Type
12.2.33 on page 676	Plasma Treater – Gas Delivery System
12.2.34 on page 680	Machine Safety Equipment / Traverse Lockout Isolator
12.2.35 on page 682	Machine Water Services
12.2.36 on page 685	Chamber Viewing Windows
12.2.37 on page 687	Electrical Lead Through - Vacuum
12.2.38 on page 688	Process Gas Wedge – Automatic System
12.2.39 on page 690	Siemens AC Vector Drive Motors
12.2.40 on page 691	Winding System Drive Belt Access Covers/Guards
12.2.41 on page 698	Bowed Spreader Roller - Kickert
12.2.42 on page 699	Winding Cart Rotary Vacuum Lead Throughs – High Speed
12.2.43 on page 700	Plasma System Power Supply
12.2.44 on page 703	Plasma High Voltage Connectors
12.2.45 on page 704	Unwind Chucks Sidelay - Screw Jack
12.2.46 on page 708	Winding mechanism Web Threader System
12.2.47 on page 709	Gearboxes

12.2.1 BUSH 630 C Version Screw Pump

Supplier's NOTE: The maintenance intervals depend on the individual operating conditions. The intervals given below should be considered as initial guidelines, which should be shortened or extended as appropriate. In particularly heavy-duty operation such as high dust loads in the environment or in the process gas, it can become necessary to shorten the maintenance intervals significantly.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the pump temperature is less than 65 C, if running warm, check the water flow to the heat exchanger
Figure 12.2-1 Example of a technician visually checking the pump temperature	 
Weekly	Check the pump for no indication of leakage of the internal coolant through the pressure relief.
Figure 12.2-2 Example of a technician visually checking for leakage	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the pump s PT100 sensor condition and cabling
Figure 12.2-3 Example of a technician visually checking for damage	
Figure 12.2-4 Example of a technician visually checking for fully engaged plugs	
Weekly	Check the temperature of the cooling fluid (see “Checking the Cooling liquid” in the vendor manual).
Weekly	Check the cooling water flow (see “Checking the cooling water” ” in the vendor manual) Record the flow, if a flow meter is installed
Figure 12.2-5 Example of a technician checking the water flow	
Weekly	Check the pump for oil leaks and repair as necessary.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Visually Check the pump oil level, in accordance with sheet attached to the pump. If sheet not available, Refer to the vendor manual.
Figure 12.2-6 Example of a technician checking oil condition & level – drive end	
Figure 12.2-7 Example of a technician checking oil condition & level – gear end	
Weekly	Check pump area for water leaks and repair as necessary.
Figure 12.2-8 Example of a technician visually checking for water leaks	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check that air is being fed into the Bush pumps through the dilution port. Placing a hand around the filter, when the pump is running will allow the technician to feel air being drawn through it. Ensure that the filter is clean. Replace if necessary.
Figure 12.2-9 Example of a technician checking air is being fed into the Bush pumps through the dilution port	
Figure 12.2-10 Example of a technician removing and checking condition of the filter	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Make visual inspection of the pump.
Monthly	In case of operation in a dusty environment, make sure that the operating room is clean and free of dust. Clean the room if necessary
Monthly	If the discharge is equipped with a silencer, bleed the condensation of the silencer through the purge system
Monthly	Check the level of the internal cooling liquid (see – Checking the cooling liquid in the vendors manual)

Maintenance Requirement	Equipment & Task to Be Performed
Bi-annual	Check electrical connections at the motor
Figure 12.2-11 Check motor connections for tightness	

Maintenance Requirement	Equipment & Task to Be Performed
Bi- annual	Replace the Oil in the pump – see supplier manual
Bi- annual	Replace the cooling medium in the pump– see supplier manual
Bi- annual	Clean/replace any inline filter unit in the water supply (if fitted). This may include back flushing of the external heat exchanger on the primary side

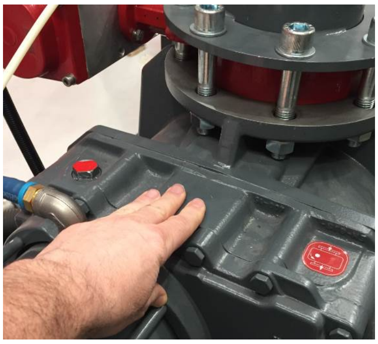
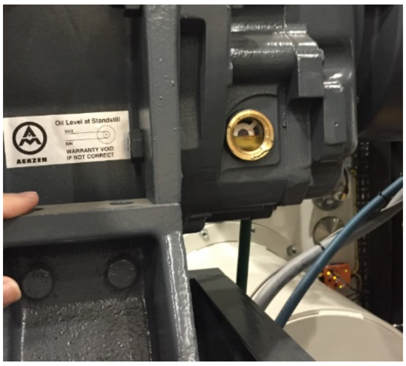
Maintenance Requirement	Equipment & Task to Be Performed
Annual	Check the vacuum inlet pipework and discharge exhaust lines and clean or replace them if necessary.

Maintenance Requirement	Equipment & Task to Be Performed
2 Years	Check the seals and replace them if necessary.
2 Years	Check the external look of the mechanical seals. If they are worn or damaged, replace them.

Maintenance Requirement	Equipment & Task to Be Performed
4 Years or 16,000 operating hours	Have major overhaul carried out by supplier (Recommended by supplier)

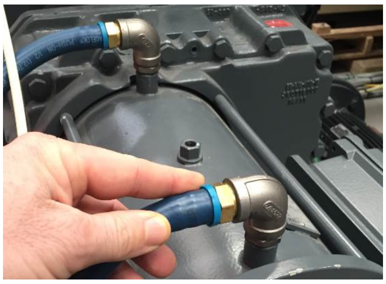
12.2.2 Mechanical Booster Pumps

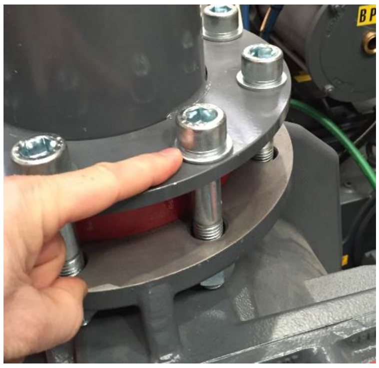
(Refer to the supplier’s manual)

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check for abnormal mechanical noises and temperature rise (hot spots- discolouration)
Figure 12.2-12 Example of a technician checking for abnormalities	
Daily	Check gear case end of the pump oil levels as per the label attached to the pump, with the pump stopped. Excess losses should be investigated.
Figure 12.2-13 Example of a technician checking the oil level on the gear case end of the Aerzen CM2000 booster pump	
Figure 12.2-14 Example of a technician checking the oil level on the gear case end of the Aerzen CM8000 booster pump	
Daily	Check drive case end of the pump oil levels as per the label attached to the pump, with the pump stopped. Excess losses should be investigated.

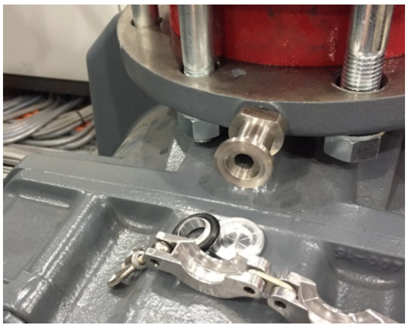
Caution !

Overfilling of oil, or use of incorrect oil grades, can lead to serious pump damage

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check the drive motor case water lines for indication of leakage and repair as necessary.
Figure 12.2-15 Example of a technician checking the condition of water lines on the Aerzen CM2000 booster pump	

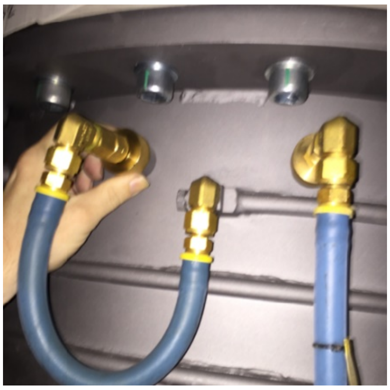
Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check locating and pipework bolts for tightness
Figure 12.2-16 Example of a technician checking the pipework bolts	
Monthly	Check the condition of the CM2000 booster pump inlet filters and clean as necessary, as described in the cleaning section of this manual.

	**Caution !**
	Ensure that the vacuum pumps are electrically isolated and the pipelines either side of the inlet and outlet of the pump are vented to atmosphere, by removing test port fittings

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Replace the oil in both sides of the pump. Ensure pipe work has been vented, both sides of the pump. Follow the guidance in the vendor manual.
Figure 12.2-17 Example of a technician removing the test port blanks to allow the pump inlet to be vented	 

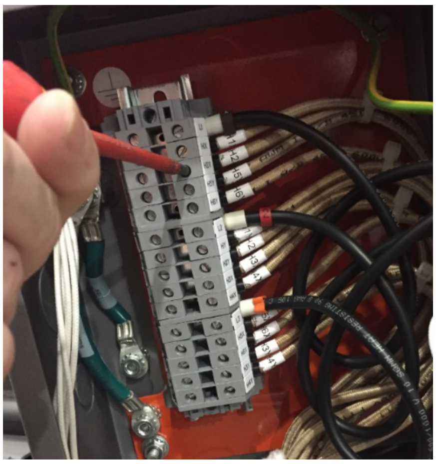
Maintenance Requirement	Equipment & Task to Be Performed
2 Years	Overhaul the pump completely- manufacturer recommendation

12.2.3 Varian Diffusion Pumps

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check for water leaks on all hoses
Figure 12.2-18 Example of a technician checking for water leaks	
Figure 12.2-19 Example of a technician checking for water leaks	
Figure 12.2-20 Example of a technician checking for water leaks	
Weekly	Check the oil level and for dis-colourment. Dirty oil should be replaced. Check the pump fluid level is at either the COLD or HOT markers on the pump sight glass
Figure 12.2-21 Example of a technician checking the pump oil level and condition	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the cooling water temperature leaving the pump is just less than 130 º F (65 ºC) at the foreline return hose
Figure 12.2-22 Example of a technician checking the water temperature at the foreline	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the pump fluid level is at either the COLD or HOT markers on the pump sight glass
Monthly	Visually check for indication of external corrosion caused by pumps being operated with the water supply less than the dew point. Correct water temperature control to stop this problem.
Monthly	Check that the heater currents are equal on all three phases. They should be balanced and agree to the level in the vendor manual
Figure 12.2-23 Example of a technician checking that the pump heater currents are equal on 3 phases	

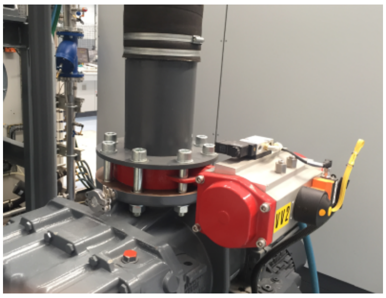
Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Inspect heater connections with the pump switched off and cold in the terminal box.
Figure 12.2-24 Example of a technician checking pump heater terminal connections	
Bi-Annual	Check the pump for its ultimate pressure

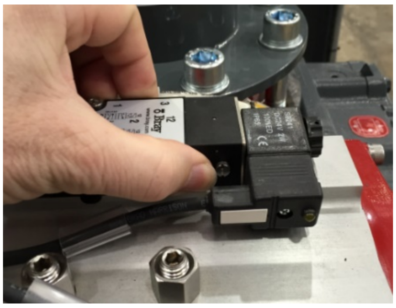
Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check the pump upper and lower thermal switches for indication of corrosion. Replace if corroded
Figure 12.2-25 Example of a technician checking pump upper (water) thermostat	
Figure 12.2-26 The location of the lower thermal switch where the hole is in the bottom plate	
Bi Annual	Check the fill and drain plugs for indication of seal failure, as indicated by silicone oil
Figure 12.2-27 Example of a technician checking for indication of leakage	
Bi-Annual	Release a water coupling line at any point in the cooling circuit and check the inside bore of the pipe for sediment build up. Flush the pump with a suitable cleaner to remove.

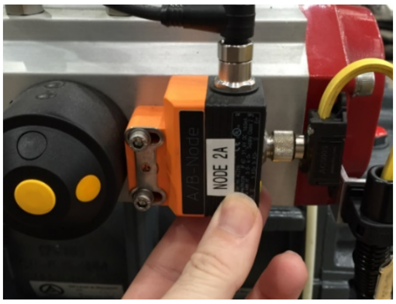
Maintenance Requirement	Equipment & Task to Be Performed
Annual	Remove pump from the machine and clean the pump internal components, check for mechanical damage, repair and replenish the oil. Install Varian preventative maintenance o-ring kit.

12.2.4 Vacuum Pipework Isolation Valves

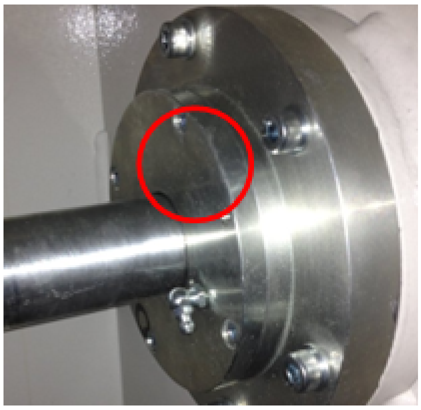
	**Warning!**
	The pipe work either side of any vacuum isolation valves should be at atmospheric pressure during maintenance. Ports are included to allow for the venting of the pipe work to atmosphere.

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check for pneumatic air leaks and solve any found.
Figure 12.2-28 Manually Check The Operations Of The Valves	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the operation of the position magnetic proximity switches, by using the mechanical override built into the pneumatic spool valve. Ensure that atmospheric pressure is on both sides of the valve, beforehand.
Figure 12.2-29 Example of a technician checking the operation of the valves	
Monthly	Check that the location bolts/nuts are tight
Figure 12.2-30 Example of a technician checking the tightness of the location bolts	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that the valve position interface unit is located correctly
Figure 12.2-31 Example of a technician checking the tightness and location of the position sensor	

12.2.5 High Vacuum Valve (VV7)

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Grease the rotary leadthroughs, with Mobil Supertemp SHC32 grease
Figure 12.2-32 The grease point on the VV7 high vacuum valve vacuum leadthrough	 
Monthly	Check that the external ETP bush has not moved. Witness marks are provided
Figure 12.2-33 The rotary leadthrough